

ELECTRONIC PRODUCTS DATASET REPORT (FAWZ ELECTRONICS)

1. Title and Data Collection Method

Title

Electronic Products Dataset Collected from Fawz Electronics

Data Collection Method

This dataset was collected manually from Fawz Electronics, a local electronics shop. The data was gathered through direct observation of available electronic products in the store. Each product's information such as name, category, quantity in stock, warranty period, brand, and price was recorded carefully.

The purpose of collecting this dataset is to understand the fundamentals of Data Science, including data collection, dataset structure, and preparation for machine learning tasks.

2. Description of Features and Label

This dataset contains information about electronic products sold in Fawz Electronics.

Features (Input Variables - X)

The following are the features used in the dataset:

- Product_Name: The name of the electronic product.
- Category: The type of product (Laptop, Smartphone, Printer, etc.).
- Quantity: The number of items available in stock.
- Warranty: The warranty period provided for the product.
- Brand: The manufacturer or company of the product.

Label (Output Variable - y)

- Price: The selling price of each product.

The label (Price) is the target variable that can be predicted using the features above.

3. Dataset Structure

The dataset contains:

- Number of Rows (Samples): 50
- Number of Columns: 6

Columns:

1. Product_Name
2. Category
3. Quantity
4. Warranty
5. Brand
6. Price

Sample Dataset (First 5 Rows)

Product_Name	Category	Quantity	Warranty	Brand	Price
Dell Inspiron 15	Laptop	10	1 Year	Dell	500
HP ProBook 450	Laptop	9	1 Year	HP	600
Lenovo ThinkPad E14	Laptop	6	1 Year	Lenovo	680
Samsung Galaxy A15	Smartphone	25	1 Year	Samsung	180
iPhone 14	Smartphone	5	1 Year	Apple	850

4. Data Quality Issues

The dataset is generally clean and well-structured. No missing values were observed in any of the columns, including Quantity, Price, and Warranty.

However, in real-world datasets, the following issues may occur:

- Typing errors during manual data entry (e.g., incorrect product names).
- Inconsistent formatting of categorical data (e.g., Laptop vs laptop).

- Possible duplicate entries of the same product.
- Variations in price due to market changes over time.
- Warranty values may not always be standardized across different products.

In this dataset, no missing values or major inconsistencies were found.

5. Type of Learning Problem

This dataset represents a **Supervised Learning** problem because it contains a labeled output variable (Price).

The model can learn from the features (Product_Name, Category, Quantity, Warranty, Brand) to predict the Price of products.

6. Use Case in Data Science and Machine Learning

This dataset can be used for a regression machine learning task where the goal is to predict product prices based on product features.

Possible Applications:

- Price prediction for new electronic products.
- Business decision-making for pricing strategies.
- Inventory and sales analysis in retail stores.

Data Science Lifecycle Stage:

This dataset belongs to the **Data Collection stage** of the Data Science lifecycle. After this stage, the data will be cleaned, analyzed, modeled, and evaluated to extract meaningful insights.

Conclusion

This project demonstrates the basics of data collection from a real-world environment. It helps in understanding dataset structure, features, labels, and the importance of data quality in Machine Learning.